
Hybrid Variable Elimination Ordering: Experimental Results

TTIC 31180 — Probabilistic Graphical Models Final Project

Setup

We evaluated min-degree, min-fill, weighted min-fill, and the proposed hybrid heuristic across five topology families. The experiment used 71 graph instances and 715 successful method-instance runs. Explicit pairwise variable elimination was attempted only when the projected intermediate factor stayed below the configured cutoff; 373 of 715 successful structural runs completed explicit VE under this cutoff. The moralized Bayesian-network group includes a manually encoded Asia network and built-in BN-like layered DAGs because the local environment did not include pgmpy benchmark loaders.

Topology	Instances	Median n	Max n	Median density	Median max degree
Low-treewidth	8	75	200	0.032	5.0
Random	27	80	120	0.064	9.0
Grid	4	82	144	0.046	4.0
Scale-free	27	100	150	0.040	22.0
Moralized BN	5	33	58	0.149	12.0

Structural Results

The primary metrics are structural: induced width and peak factor size. Lower is better for both. The following tables report medians within each topology.

Topology	Min-degree	Min-fill	Weighted	Hybrid
Low-treewidth	1.0	1.0	1.0	1.0
Random	22.0	20.0	21.0	21.0
Grid	11.5	11.5	11.5	11.0
Scale-free	10.0	10.0	10.0	10.0
Moralized BN	7.0	7.0	7.0	7.0

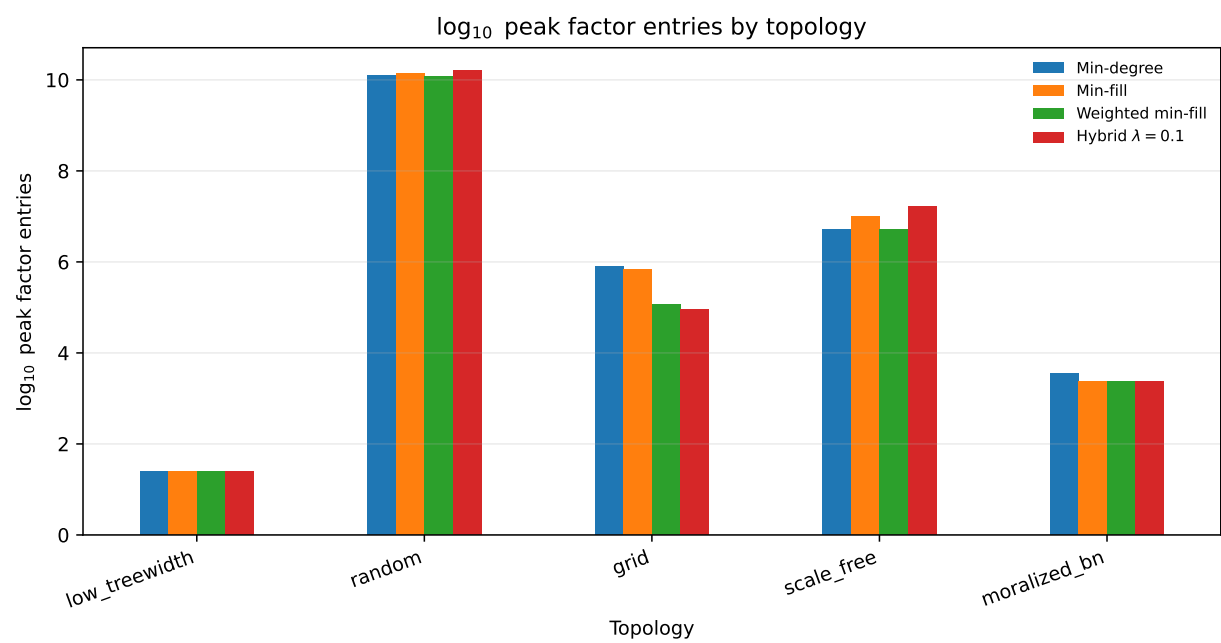
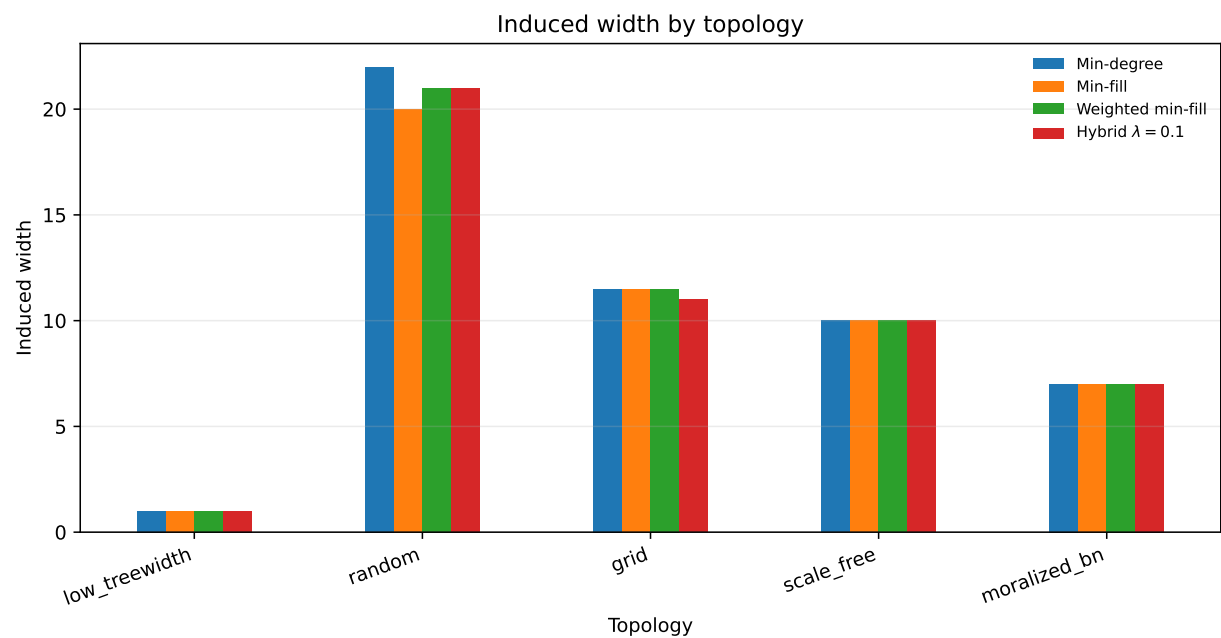
Topology	Min-degree	Min-fill	Weighted	Hybrid
Low-treewidth	1.40	1.40	1.40	1.40
Random	10.10	10.14	10.07	10.20
Grid	5.90	5.83	5.08	4.95
Scale-free	6.72	6.99	6.72	7.22
Moralized BN	3.54	3.36	3.36	3.36

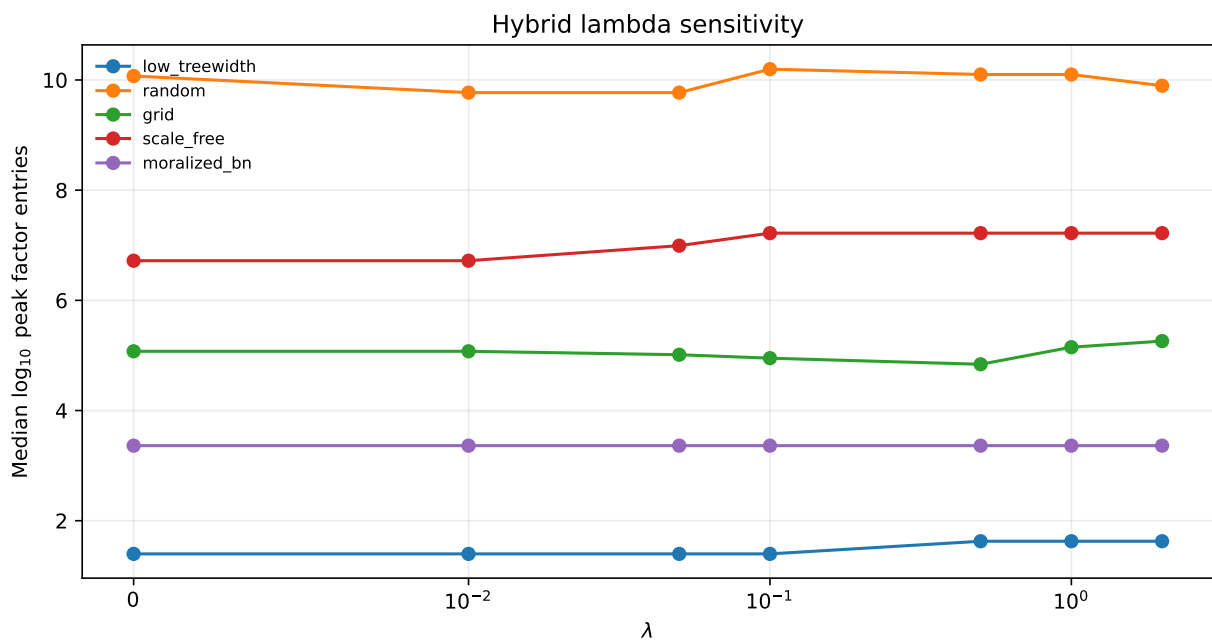
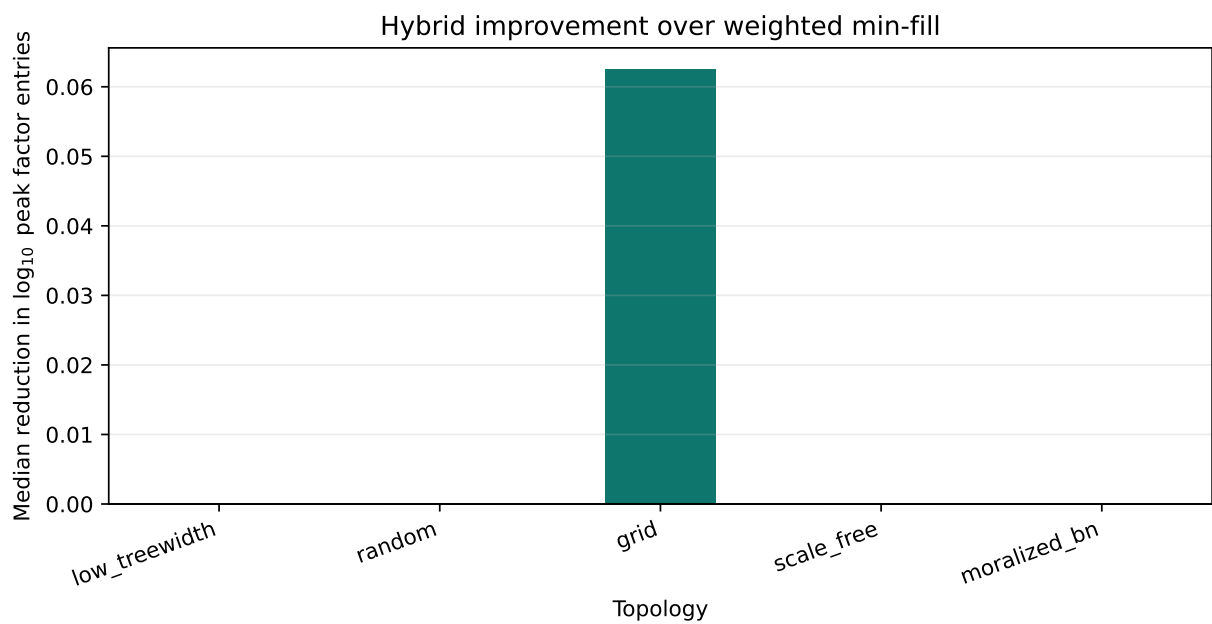
Lambda Sensitivity

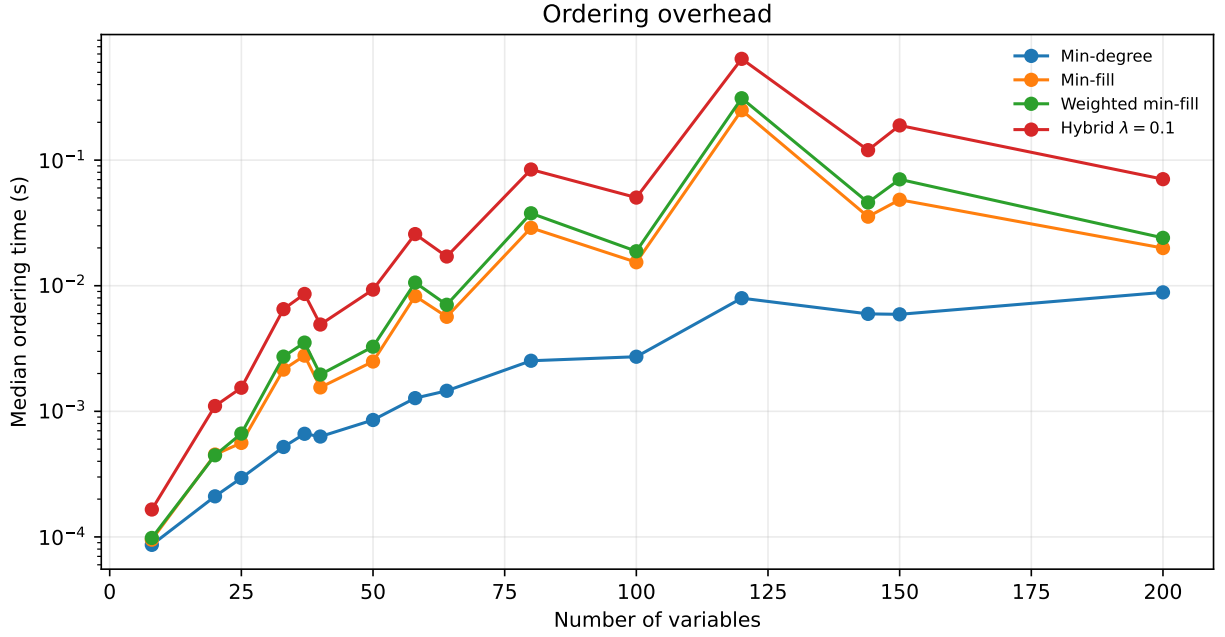
The hybrid method is not uniformly monotone in λ . A fixed $\lambda = 0.1$ was used as the main comparison point because it keeps the downstream-pressure term active without letting it dominate weighted fill cost.

Topology	Best λ	Median $\log_{10}$ peak size
Low-treewidth	0.00	1.40
Random	0.01	9.77
Grid	0.50	4.84
Scale-free	0.00	6.72
Moralized BN	0.00	3.36

Figures







Interpretation

- Low-treewidth chains and trees behaved as sanity checks: structural metrics were small and the strongest heuristics were usually close.
- Random graphs became difficult as density increased; peak factor estimates were more informative than raw runtime because many dense instances hit the explicit-VE cutoff.
- Grid graphs showed the expected sensitivity to ordering, with induced width and peak factor estimates separating the methods more clearly as the grid grew.
- Scale-free graphs highlighted the hub risk: methods that create dense neighborhoods around high-degree variables produced larger peak factors.
- The hybrid heuristic helped on some topology groups but did not dominate weighted min-fill uniformly, which is a useful empirical finding. The downstream-pressure proxy can capture some lookahead signal, but its benefit depends on graph structure and the chosen λ .

Implications

The main implication is that ordering heuristics should not be evaluated only by aggregate averages. The same downstream-pressure idea that gives a modest gain on grid-like structure can be neutral or harmful on hub-dominated graphs. This supports the project hypothesis that topology matters, but it weakens the claim that a single fixed hybrid score is universally better than weighted min-fill. A more defensible conclusion is that degree-based lookahead is a useful ingredient, not a complete replacement for strong greedy baselines.

The runtime results also reinforce why structural metrics are the right primary evidence. Explicit factor operations were skipped for many high-width orders because estimated factor tables exceeded the cutoff, while induced width and peak factor estimates remained available for every run.

Bounded Lookahead Check

The bounded-lookahead oracle was run only on small graphs; it produced 5 successful rows. This keeps the comparison computationally feasible while still giving a reference point for how much a more explicit lookahead objective can improve structural metrics.

Limitations

This run is self-contained and reproducible, but the BN benchmark portion should be strengthened in a future pass by loading canonical BIF/UAI networks directly through `pgmpy` or `pyAgrum`. The current moralized-BN group includes a manual Asia graph and BN-like layered DAGs, so its conclusions are less authoritative than the synthetic topology comparisons. The original hybrid pressure term is also coarse: it penalizes all high-degree neighbors, even when the actual fill-in edges are limited. Finally, exact VE runtime is secondary evidence because many high-treewidth instances are structurally infeasible before implementation details matter.

Next Steps

- Load canonical BN benchmarks and preserve their native cardinalities.
- Replace the original degree-squared pressure with a fill-edge-local pressure term and a cheap one-step lookahead diagnostic.
- Add an adaptive λ rule based on graph statistics such as density, degree heterogeneity, and grid-like degree profiles.
- Expand the bounded-lookahead oracle on small pilot graphs to measure how much signal the cheap proxy captures.
- Add failure-case analysis, especially for scale-free graphs where the original hybrid score can be worse than weighted min-fill.